Program : **Diploma in Cyber Forensics And Information security**	
Course Code : **4282**	Course Title: **Cyber Forensics and Security Threats**
Semester :**4**	Credits: **4**
Course Category: **Program Core**	
Periods per week: 4 **(L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- To introduce computer forensics concepts, as well as techniques for identifying, collecting, and preserving digital evidence consistent with industry standards and best practices.
- An in-depth study of each phase involved in a forensics investigation process.
- To understand system and network security threats and the defenses against them.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic knowledge in Computer Systems		Introduction to IT Systems Lab	I

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand basic ideas of Computer Forensics and the forensic investigations	14	Understanding
CO2	Outline how to acquire data from devices and the theory behind various types of data acquisitions	15	Understanding
CO3	Describe how to perform analysis and validation of data, prepare reports and expert testimony	14	Understanding
CO4	Summarize various security threats	15	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2		2				
CO2	2		2				
CO3	-					1	
CO4	1	3	3			1	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Understand basic ideas of Computer Forensics and the forensic investigations**		
M1.01	Explain basic concepts of Cyber Forensics	2	Understanding
M1.02	Explain case law	1	Understanding
M1.03	Explain developing computer forensics resources	2	Understanding
M1.04	Summarize Computer investigations-systematic approach, and maintaining professional conduct	2	Understanding
M1.05	Explain procedures for corporate high tech investigations	3	Understanding
M1.06	Explain types of data recovery workstations and software	2	Understanding
M1.07	Explain Requirements for forensic lab	2	Understanding
Contents: Introduction to Computer Forensics - history of computer forensics, understanding case law, preparing for computer investigations, understanding corporate investigations, maintaining professional conduct. Understanding Computer Investigations - Preparing a computer investigation, taking a systematic approach, procedures for corporate high tech investigations, understanding data recovery workstations and software, conducting an investigation, completing the case. Requirements for forensic lab certification-determining the physical requirements for a computer forensics lab, selecting a basic forensic workstation.			

CO2	**Outline how to acquire data from devices and the theory behind various types of data acquisitions**		
M2.01	Summarize storage format	2	Understanding
M2.02	Explain Types of image acquisition	1	Understanding
M2.03	Describe validating data acquisitions	2	Understanding
M2.04	Explain RAID data acquisition, remote network acquisition,other forensic acquisition tool.	1	Understanding
M2.05	Explain various steps for processing crime and incident scene	5	Understanding
M2.06	Describe storing of digital evidence and obtaining a digital hash.	3	Understanding
	Series Test – I	1	
Contents : Data Acquisition - storage formats for digital evidence, determining the best acquisition method, contingency planning for image acquisitions, using acquisition tools, validating data acquisitions, performing RAID data acquisitions, using remote network acquisition tools, using other forensic acquisition tools.Processing Crime and Incident Scene - identifying digital evidence, collecting evidence in private sector incident scenes, processing law enforcement crime scenes, preparing for a search, securing a computer incident or crime scene,Seizing digital evidence at the scene, storing digital evidence, obtaining a digital hash.			
CO3	Describe how to perform analysis and validation of data prepare reports and expert testimony		
M3.01	Explain analysis and validation	3	Understanding
M3.02	Compare types of data-hiding techniques	3	Understanding
M3.03	Explain basic concepts of report writing, report findings with forensic tool	4	Understanding
M3.04	Describe generation report findings with forensics software tools	4	Understanding
Contents : Analysis and validation-determining what data to collect and analyse, validating forensic data, addressing data-hiding techniques, performing remote acquisitions. Report writing for high tech investigations – importance of reports, guidelines for writing, generating report findings with forensics software tools.			

CO4	**Summarize various security threats**		
M4.01	Explain Security threats, sources , Email threats, Web-threats	3	Understanding
M4.02	Summarize intruders hackers, insider threats, cyber crimes	3	Understanding
M4.03	Compare Active and passive attacks	2	Understanding
M4.04	Compare Worms –Virus – Spam’s – Ad ware - Spy ware – Trojan	2	Understanding

M4.05	Summarize IP, Spoofing - ARP spoofing - Session Hijacking	2	Understanding
M4.06	Compare Internal threats Environmental threats	2	Understanding
	Series Test – II	1	
Security threats: Introduction: Security threats - Sources of security threats- Motives - Target Assets and vulnerabilities – Consequences of threats- E-mail threats - Web-threats - Intruders and Hackers, Insider threats, Cyber crimes. Network Threats: Active/ Passive – Interference – Interception –Impersonation – Worms –Virus – Spam's – Ad ware - Spy ware – Trojans and covert channels –Backdoors – Bots – IP Spoofing - ARP spoofing - Session Hijacking - Sabotage-Internal threats Environmental threats - Threats to Server security			

Text / Reference

T/R	Book Title/Author
T1	Guide to computer forensics and investigation 4th edition by Amelia Philip, Bill Nelsonand Christopher Steuart.
R1	David Benton and Frank Grindstaff ,Practical guide to Computer Forensics - Book Surge Publishing,2006, ISBN-10: 1419623877
R2	Christopher L.T Brown Charles ,Computer Evidence: Collection & Preservation - River Media publishing, Edition 1, 2005 ISBN-10: 1584504056
R3	Elizabeth Bauchner, Computer Investigation (Forensics, the Science of crime-solving) – Mason Crest Publishers, 2005 ISBN-10: 1422200353
R4	Keith J. Jones, Richard Bejtlich and Curtis W. Rose, Real Digital Forensics - Addison-Wesley publishers, 2005 ISBN-10: 0321240693
R5	Swiderski, Frank and Syndex, "Threat Modeling", Microsoft Press, 2004. 2.
R6	William Stallings and Lawrie Brown, "Computer Security: Principles and Practice", Prentice Hall

Online Resources

Sl No	Website Link
1	https://www.w3schools.in/cyber-security/cyber-forensics-and-incident-handling/
2	https://www.geeksforgeeks.org/cyber-forensics